

Cake Cutting

Properties of the Cut and Choose Protocol

Today, we'll focus primarily on fairness properties.

Suppose two people need to split a heterogeneous and divisible good. The usual euphemism in the fair division literature is that of cutting a cake. More practically, the good could be an estate (e.g., in a divorce settlement) or processing time on a computer cluster (perhaps with some times of the day more valuable than others).

Why not just split the good 50/50? This makes sense when the good is homogeneous, but it's not clear what this means with a heterogeneous good. A player may value a part of the good much more than another, and different players can have different opinions about which parts are the most valuable.

We all know a reasonable protocol for two-person cake-cutting.

The Cut and Choose Protocol

1. Player 1 splits the good into two pieces A and B , such that the player's value for each is exactly half that of the entire good.
2. Player 2 picks whichever of A, B she likes better.

The description above is the intended behavior of the players in the protocol—we'll talk shortly about whether or not they are incentivized to follow this behavior. We've been led all our lives to believe that this is a "fair" protocol. But is it? How would we formally argue one way or the other?

Here's the formal model. The good, or "cake," is the unit interval $[0, 1]$. (Yes, it's a weird-looking cake.) Each player i has a valuation function v_i , which specifies the value $v_i(S)$ to i of a given subset S of the cake. Values are nonnegative, and we assume that $v_i([0, x])$ is continuous in x , so a cut giving player i half her value exists. Pieces may be unions of intervals, and we allocate the entire cake.

1. We normalize each v_i such that $v_i([0, 1]) = 1$, their value for the whole cake. This is more or less without loss of generality, by scaling.
2. v_i is additive on disjoint subsets. That is, if $A, B \subseteq [0, 1]$ are disjoint, then

$$v_i(A) + v_i(B) = v_i(A \cup B).$$

Exercise. Player 1 values the cake uniformly. Player 2 values only the left quarter, uniformly within that quarter. Run cut-and-choose with player 1 cutting at $1/2$. What does each player receive, and how much does she value it?

Is the cut and choose protocol strategyproof?

Here, ask whether either player can gain by departing from the prescribed behavior while the other follows it.

An allocation is *Pareto optimal* if there is no other allocation that makes both players weakly better off and at least one strictly better off.

Is the cut and choose protocol guaranteed to produce a Pareto optimal solution (assuming both players behave as intended)?

Consider four equal quarters. Player 1 values only quarters 1 and 3, at $1/2$ each; player 2 values only quarters 2 and 4, at $1/2$ each. Compare cutting into two halves with another allocation.

Can we come up with a protocol that is both Pareto optimal and strategyproof for both players?

For this question, suppose both players value every set of positive length strictly positively.

What about “fairness?” Equally happy?

Does cut-and-choose give both players the same value for their own pieces?

One definition of “fairness” is that each player receives at least her fair share, at least from her perspective.

Definition 1. An allocation $A_1, \dots, A_n$ of the entire cake to n players is *proportional* if

$$v_i(A_i) \geq$$

for each player i .

A second definition is that no player wants to trade places with the other player.

Definition 2. An allocation $A_1, \dots, A_n$ is *envy-free* if

$$v_i(A_i) \geq$$

for each pair i, j .

This compares both pieces using the same player’s valuation.

Does cut-and-choose satisfy these properties? How are they related for general n , and for two players?

Assume both players follow the protocol.

Fair Division of Indivisible Goods

Exercise. Two players both value a single indivisible good at 1. Allocate the good. Can the allocation be envy-free or proportional?

Definition 3. We have a set $N = [n]$ of players and a set $\mathcal{M}$ of m goods. Each player i has a nonnegative, additive valuation:

$$v_i(S) = \sum_{g \in S} v_i(g).$$

An allocation $A = (A_1, \dots, A_n)$ is a partition of $\mathcal{M}$ into disjoint bundles, one per player. Empty bundles are allowed; all goods are allocated.

For the numerical examples, we leave the values unnormalized. Proportionality then asks for $v_i(A_i) \geq v_i(\mathcal{M})/n$.

Notions of Envy Freeness

Our definition of envy-freeness is the same as before. Since it is not always possible to achieve it, we relax the condition.

Definition 4. For an allocation $A = (A_1, \dots, A_n)$:

1. The allocation is *envy-free up to one good* (EF1) if for every i, j with $A_j \neq \emptyset$,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for some } g \in A_j.$$

2. The allocation is *envy-free up to any good* (EFX) if for every i, j ,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for every } g \in A_j.$$

An empty bundle is never envied. Removing a good is only a comparison: the actual allocation still contains all the goods. Here EFX uses every good, including zero-valued goods.

How do the quantifiers differ? Which condition is stronger?

Two-Player Cut and Choose

Exercise. When can we achieve an EF1 allocation? Try $n = 2$ players, any m goods, and additive valuations. [*Hint:* Try something like cut-and-choose.] Use the following values to illustrate your rule.

Player	A	B	C
1	10	9	2
2	1	8	4

Round Robin

The Round Robin Mechanism. First, set an arbitrary order of the players, denoted $1, 2, \dots, n$. Until all the goods are allocated, go through the players in this order repeatedly. On player i 's turn, she chooses her most valuable remaining good. Ties may be broken arbitrarily.

Theorem 1. *For nonnegative additive valuations, round robin produces an EF1 allocation.*

Let $g_{i,k}$ denote the good picked by player i in round k , when that pick exists. Compare two players i, j , evaluating both players' picks using v_i .

Proof. **Case $i < j$.** Draw the pick order. Which good was still available when i made her k th pick?

Case $i > j$. Draw the pick order. Which good should be removed from j 's bundle, and how do the remaining picks compare?

How Unfair Can Round Robin Be?

Exercise. There are $n \geq 2$ players with identical additive values: $n - 1$ goods worth n each and n goods worth 1 each. Run round robin in the order $1, \dots, n$. Compare with another allocation. Is the round-robin outcome EFX?

Maximin Share

Suppose a player could partition the goods into n bundles, but had to take her least valuable bundle. What value could she guarantee herself?

Definition 5. The *maximin share* of player i is

$$\text{MMS}_i = \max_{(B_1, \dots, B_n) \in \Pi_n(\mathcal{M})} \min_{1 \leq j \leq n} v_i(B_j),$$

where $\Pi_n(\mathcal{M})$ is the set of partitions into n bundles, possibly empty. An allocation A is *MMS-fair* if $v_i(A_i) \geq \text{MMS}_i$ for every player i .

Whose valuation is used for all n bundles in this definition? How does MMS_i compare with the proportional share $v_i(\mathcal{M})/n$?

Exercise. In the example with $n - 1$ goods worth n and n goods worth 1, compute each player's MMS. What fraction of her MMS does the last player receive under round robin?

Optional: Two-Player MMS

Exercise. With two players and additive valuations, player 1 chooses a partition attaining her MMS, and player 2 chooses her preferred bundle. Prove that this is MMS-fair. Is the earlier greedy EF1 partition necessarily an MMS partition?